dynamical-collapse models, the most famous of which is *GRW theory*, after its inventors Giancarlo Ghirardi, Alberto Rimini, and Tullio Weber.

Envision an electron in free space, not bound to any atomic nucleus. According to the Schrödinger equation, the natural evolution of such a particle is for its wave function to spread out and become increasingly diffuse. To this picture, GRW adds a postulate that says at every moment there is some probability that the wave function will change radically and instantaneously. The peak of the new wave function is itself chosen from a probability distribution, the same one that we would have used to predict the position we would measure for the electron according to its original wave function. The new wave function is strongly concentrated around this central point, so that the particle is now essentially in one location as far as we macroscopic observers are concerned. Wave function collapses in GRW are real and random, not induced by measurements.

GRW theory is not some nebulous "interpretation" of quantum mechanics; it is a brand-new physical theory, with different dynamics. In fact, the theory postulates two new constants of nature: the width of the newly localized wave function, and the probability per second that the dynamical collapse will occur. Realistic values for these parameters are perhaps 10^{-5} centimeters for the width, and 10^{-16} for the probability of collapse per second. A typical electron therefore evolves for 10^{16} seconds before its wave function spontaneously collapses. That's about 300 million years. So in the 14-billion-year lifetime of the observable universe, most electrons (or other particles) localize only a handful of times.

That's a feature of the theory, not a bug. If you're going to go messing around with the Schrödinger equation, you had better do it in such a way as to not ruin all of the wonderful successes of conventional quantum mechanics. We do quantum experiments all the time with single particles or collections of a few particles. It would be disastrous if the wave functions of those particles kept spontaneously collapsing on us. If there is a truly random element in the evolution of quantum systems, it should be incredibly rare for individual particles.

Then how does such a mild alteration of the theory manage to get rid of macroscopic superpositions? Entanglement comes to the rescue, much as it did with decoherence in Many-Worlds.

Consider measuring the spin of an electron. As we pass it through a Stern-Gerlach magnet, the wave function of the electron evolves into a